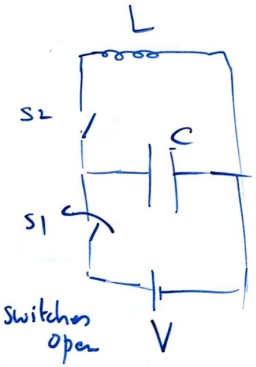
7.7.26

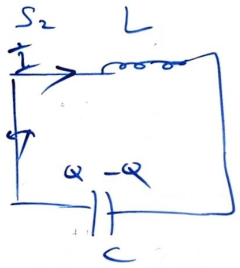
LC - circuit



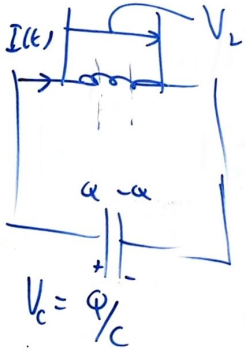
0. All switches open
1. Close S_1, charge C
$Q_0 = CV$
2. Open S_1



3. Close S_2



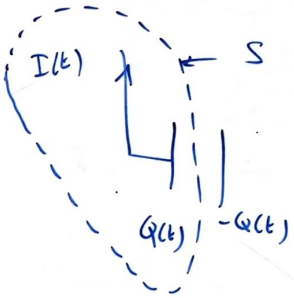
$Q(0) = Q_0 = CV$



$V_C = Q/C$

$V_L = V_C$

$L\frac{dI}{dt} = \frac{Q}{C}$



t: tot. charge $= Q(t)$

$t+\Delta t$: " " $= Q(t+\Delta t) + I\,\Delta t$

$Q(t+\Delta t) + I\,\Delta t = Q(t)$

$\frac{Q(t+\Delta t) - Q(t)}{\Delta t} = -I \Rightarrow \frac{dQ}{dt} = -I$

$L\frac{dI}{dt} = \frac{Q}{C}$

$I = -\frac{dQ}{dt}$

$\Rightarrow L\frac{d^2Q}{dt^2} = -\frac{1}{C}Q$

$$\boxed{\frac{d^2Q}{dt^2} = -\frac{1}{LC}Q}$$

$\Rightarrow \omega^2 = \frac{1}{LC}$

$\omega = \frac{1}{\sqrt{LC}}$

$T = 2\pi\sqrt{LC}$

$1\cdot\frac{d^2Q}{dt^2} = -\ldots$

$m\frac{d^2x}{dt^2} = -\ldots$

Variables	comparison
x	Q
m	L
k	$1/C$
$U = \frac{1}{2}kx^2$	$\frac{1}{2C}Q^2$
$k = \frac{1}{2}mv^2$	$\frac{1}{2}LI^2$
$= \frac{1}{2}m\left(\frac{\Delta x}{dt}\right)^2$	$= \frac{1}{2}L(\ldots$

$= \frac{Q}{C}$

$- \frac{dQ}{dt}$

$= -\frac{1}{C} Q$

$= -\frac{1}{LC} Q$

$= \frac{1}{LC}$

$\omega = \frac{1}{\sqrt{LC}}$

$T = 2\pi\sqrt{LC}$

$\int I \, dt$

$\frac{d\varphi}{dt} = -I$

$$L \frac{d^2Q}{dt^2} = -\frac{1}{C} Q$$

$$m \frac{d^2x}{dt^2} = -kx$$

Variables	Comparison
x	Q
m	L
k	$1/C$
$U = \frac{1}{2}kx^2$	$\frac{1}{2C}Q^2$
$K = \frac{1}{2}mv^2 = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2$	$\frac{1}{2}LI^2 = \frac{1}{2}L\left(\frac{dQ}{dt}\right)^2$

Averages, Avg. KE, Avg. PE

$\rightarrow x = A \sin \omega t$,

$\dot{x} = \omega A \cos \omega t$

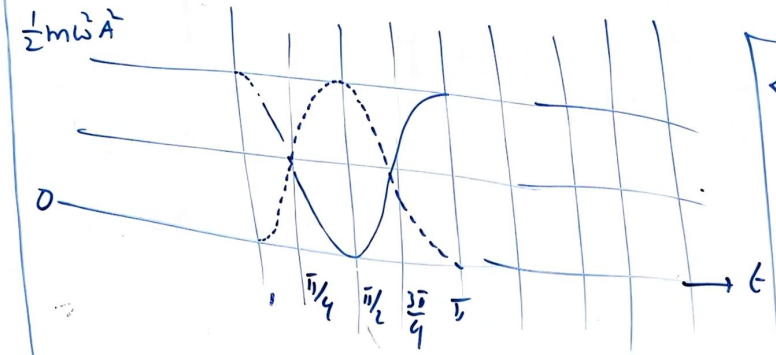
$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2 \sin^2 \omega t$

$\frac{k}{m} = \omega^2$ $\quad k = m\omega^2$

$\theta = \omega t$

$U = \frac{1}{2}m\omega^2 A^2 \sin^2 \omega t = \frac{1}{2}m\omega^2 A^2 \frac{(1 - \cos 2\theta)}{2}$

$K = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2 A^2 \cos^2 \omega t = \frac{m\omega^2 A^2}{2} \frac{(1 + \cos 2\theta)}{2}$



$\langle K \rangle = \frac{1}{4}m\omega^2 A^2$

$\langle K \rangle + \langle U \rangle = \frac{1}{2}m\omega^2 A^2$

Avg value of a function

$$\langle f(t) \rangle_T = \frac{1}{T} \int_0^T f(t) \, dt$$

if $f(t+T) = f(t)$

$$= \frac{1}{T} \int_{t_1}^{T+t_1} f(t) \, dt$$

$$\langle U \rangle = \frac{1}{T} \int_0^T \frac{1}{2}m\omega^2 A^2 \sin^2 \omega t \, dt$$

θ

$$= \frac{1}{2} \frac{1}{2\pi} \omega \cdot m\omega A^2 \int_0^{2\pi} \sin^2 \theta \, d\theta$$

$$= \frac{1}{2}m\omega^2 A^2 \frac{1}{2\pi} \int_0^{2\pi} \sin^2 \theta \, d\theta = \frac{1}{2}$$

$$\langle U \rangle = \frac{1}{4}m\omega^2 A^2$$

7.7-26

SHM in arbitrary potentials having smooth minimum.

Taylor expand around r_0

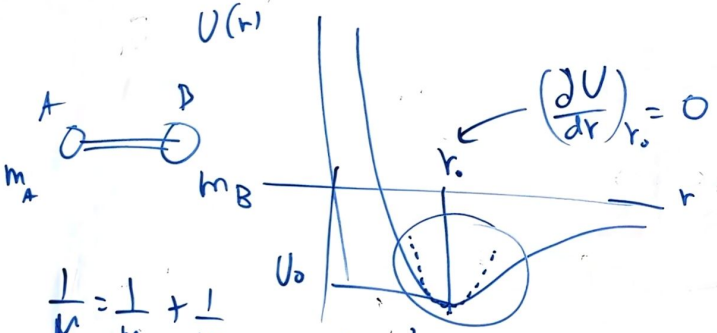
$$U(r) = U(r_0) + (r-r_0)\left.\frac{dU}{dr}\right|_{r_0} + \frac{1}{2}(r-r_0)^2\left.\frac{d^2U}{dr^2}\right|_{r_0} + \cdots$$

$\left.\frac{dU}{dr}\right|_{r_0} = 0$ by def. r_0 is equil. pt.

$\left.\frac{d^2U}{dr^2}\right|_{r_0} = k$

For small osc.

$$\approx U(r_0) + \frac{k}{2}(r-r_0)^2$$



$$\frac{1}{\mu} = \frac{1}{m_A} + \frac{1}{m_B}$$

$$\mu\frac{d^2r}{dt^2} = -\frac{dU(r)}{dr}$$

$$= -k(r-r_0) \longrightarrow$$

$$-\frac{dU}{dr} = -k(r-r_0)$$

Define $r - r_0 = x$

$$\frac{dx}{dt} = \frac{dr}{dt}, \quad \frac{d^2x}{dt^2} = \frac{d^2r}{dt^2}$$

$$\mu\frac{d^2x}{dt^2} = -kx \Rightarrow \boxed{\frac{d^2x}{dt^2} = -\frac{k}{\mu}x}$$

$$\omega = \sqrt{\frac{k}{\mu}}$$

$$\boxed{\omega = \sqrt{\frac{U''(r_0)}{\mu}}}$$

L

m

Variables

x

m

k

$U = \frac{1}{2}kx^2$

$k = \frac{1}{2}mv^2$

$= \frac{1}{2}m\left(\frac{dx}{dt}\right)^2$